

ChE 201 – Introduction to Thermodynamics –Spring 2016

Student Learning Outcome 1

Rubric for assessment problem on final exam (Problem #4)

Application of energy and entropy balances to an adiabatic air compressor

SLO 1 Performance Indicator (PI)	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject Problem 4a, 4b 10 pts	Does not understand the connection between mathematical models and the system or process to be analyzed or designed (<3 pts)	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development (3-5 pts)	Able to successfully combine mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering (6-10 pts)
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra Problem 4c 25pts	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems (<7 pts)	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving (7-14 pts)	Applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems (15-25 pts)
4. Translates academic theory into engineering applications Problem 4d 10 pts	Does not appear to grasp the connection between theory and the problem (<3 pts)	Some gaps in understanding the application of theory to the problem and expects theory to predict reality (3-5 pts)	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality (6-10 pts)

Averages:

Performance Indicators: (1) 3.00 (2) 2.62 (4) 2.33 Overall Average: 2.65

Average scores showed all 63 students were very strong in formulating the engineering system and model equations. A majority were satisfactory in the skills to solve engineering problems and translating theory into engineering applications (41/63 and 35/63).

QUESTION 4: (45 pts)

A 2.5 kg/min stream of air is compressed at steady state from an initial state of 100 kPa and 22°C to a final pressure of 800 kPa. Assume the air is an ideal gas with temperature dependent heat capacity and neglect any changes in kinetic and potential energy.

- a) (5 pts) Draw a schematic of the system and label the system boundary.
- b) (5 pts) State the engineering model
- c) (25 pts) Determine the outlet temperature and the minimum power required when:

(1) Compressor is adiabatic $T_2 = \underline{\hspace{2cm}} \text{ K}$ $\dot{W} = \underline{\hspace{2cm}}$
kW

(2) Compressor is isothermal $T_2 = \underline{\hspace{2cm}} \text{ K}$ $\dot{W} = \underline{\hspace{2cm}}$
kW

- d) (10 pts) Draw the adiabatic and isothermal compression processes on a p-v diagram.

Solution

CHE 201 Final Exam

May 7th, 2016

QUESTION 4: (45 pts)

A 2.5 kg/min stream of air is ~~adiabatically~~ compressed at steady state from an initial state of 100 kPa and 22°C to a final pressure of 800 kPa. Assume the air is an ideal gas and neglect any changes in kinetic and potential energy.

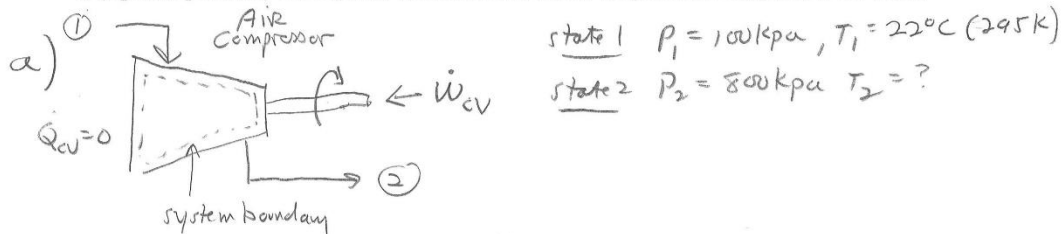
with temperature dependent
specific heat

- (5 pts) Draw a schematic of the system and label the system boundary.
- (5 pts) State the engineering model
- (25 pts) Determine the outlet temperature and the minimum power required when:

20 ~~pts~~ (1) Compressor is adiabatic $T_2 = 530.12$ K $\dot{W} = -10.00$ kW

5 ~~pts~~ (2) Compressor is isothermal $T_2 = 295$ K $\dot{W} = -7.34$ kW

- (5 pts) Draw the adiabatic and isothermal compression processes on a p-v diagram.
- (5 pts) Explain why the required work is less for one of the processes compared to the other.



b) Engineering Model

- Open system is air flowing in compressor
- $\Delta KE, \Delta PE = 0$
- Adiabatic, $\dot{Q}_{cv} = 0$ or isothermal ($\Delta T = 0$)
- shaft work is only work mode, $\dot{W}_{cv}$
- Assume air is an ideal gas, steady state

c) (1) $\dot{Q}_{cv} = 0$ 1st Law Balance: $\dot{W}_{cv} = \dot{m}(h_1 - h_2) \Rightarrow$
2nd Law Balance: $0 = \dot{m}(s_1 - s_2) + \dot{Q}_{cv}$

$h_2 = ?$ since minimum power; $\dot{Q}_{cv} = 0$ (Reversible process)
2nd Law $\rightarrow s_1 = s_2$ (Isentropic process)

$\dot{m} = 2.5 \text{ kg/min}$
 $h_1 = 295.17 \text{ kJ/kg}$
from Table A-22

c) (1) (cont.) $S_1 = S_2$ so eqn 6.20a becomes:

$$S(T_2, P_2) - S(T_1, P_1) = S^\circ(T_2) - S^\circ(T_1) - R \ln P_2/P_1 = 0$$

$$S^\circ(T_2) = S^\circ(T_1) + R \ln P_2/P_1 \quad \text{Table A-22}$$

$$S^\circ(T_2) = 1.68515 + \frac{8.314 \text{ kJ}}{\text{kmol} \cdot \text{K}} \frac{1 \text{ kmol}}{28.97 \text{ kg}} \ln \left(\frac{800 \text{ kPa}}{100 \text{ kPa}} \right) \quad S^\circ(295 \text{ K}) = 1.68515 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$$

$$S^\circ(T_2) = 1.68515 \frac{\text{kJ}}{\text{kg} \cdot \text{K}} + 0.2869 \cdot 2.079 \frac{\text{kJ}}{\text{kg} \cdot \text{K}} =$$

$$S^\circ(T_2) = 1.68515 + 0.59677 = 2.28192 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$$

$$T_2 @ S^\circ(T_2) \Rightarrow \text{Interpolate: } \frac{T_2 - 530}{(540 - 530)} = \frac{(2.28192 - 2.27967)}{(2.29906 - 2.27967)}$$

$$T_2 = (10 \text{ K}) \left(\frac{0.00225}{0.01939} \right) + 530 \text{ K} = 10(0.116) + 530$$

$$T_2 = 530.12 \text{ K}$$

$$h_2 = ? @ 530.12 \text{ K} \quad \text{Interpolate: } \frac{h_2 - 533.98}{(544.35 - 533.98)} = (0.116)$$

$$h_2 = (10.37)(0.116) + 533.98 = 1.203 + 533.98$$

$$h_2 = 535.18 \frac{\text{kJ}}{\text{kg}} \Rightarrow \dot{W} = \left(2.5 \frac{\text{kg}}{\text{min}} \right) (295.17 - 535.18) \frac{\text{kJ}}{\text{kg}}$$

$$\dot{W} = \left(2.5 \frac{\text{kg}}{\text{min}} \right) \left(-240.01 \frac{\text{kJ}}{\text{kg}} \right) \frac{1 \text{ min}}{60 \text{ sec}}$$

$$\dot{W} = \frac{-600.03 \text{ kJ}}{\text{min}} \frac{1 \text{ min}}{60 \text{ sec}} \frac{1 \text{ kW}}{1 \text{ kJ/s}}$$

$$\dot{W} = -10.00 \text{ kW}$$

c)

(2) Isothermal case: 1st Law: Ideal gas $\Delta U = 0$ so $\dot{W} = \dot{Q}$

$$T_1 = T_2 = 295 \text{ K} \quad \text{2nd Law:} \quad 0 = \frac{\dot{Q}}{T} + \dot{m}(s_1 - s_2) + \frac{\dot{Q}}{T} \frac{1}{c_v}$$

$$\Delta S = \dot{m}(s_2 - s_1) = \dot{m} \left[s^\circ(T_2) - s^\circ(T_1) - R \ln \frac{P_2}{P_1} \right]$$

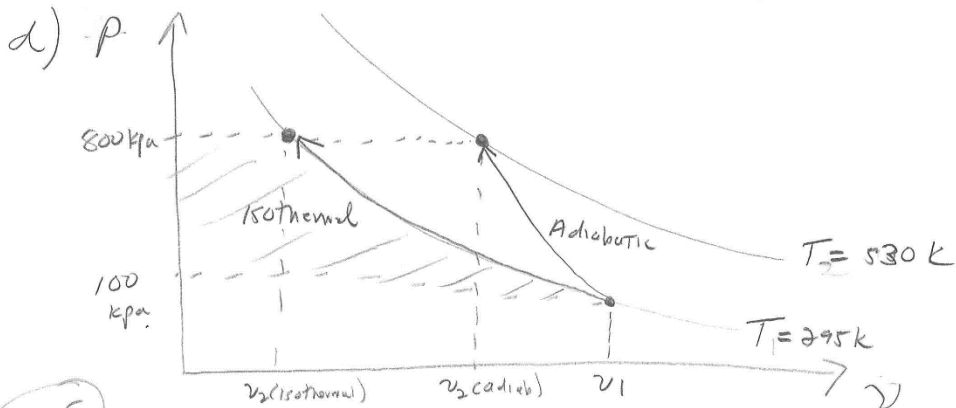
$= 0$ isothermal case

$$(s_2 - s_1) = -R \ln \frac{P_2}{P_1}$$

$$\dot{W} = \dot{Q} = T \dot{m} (-R \ln \frac{P_2}{P_1}) = (295 \text{ K}) \left(\frac{2.5 \text{ kg}}{\text{min}} \right) \left(-\frac{8.314 \text{ J}}{28.97 \text{ kg} \cdot \text{K}} \right) \ln \left(\frac{800}{100} \right)$$

$$= -211.65 \frac{\text{kJ}}{\text{min}} (2.079) = 440.12 \frac{\text{kJ}}{\text{min}} \left| \frac{1 \text{ min}}{60 \text{ sec}} \right| \frac{1 \text{ kW}}{1 \text{ kJ/s}}$$

$$\dot{W}_{\text{isotherm}} = -7.34 \text{ kW}$$



Take it

e) $W = \int v dp$ $(\int v dp)_{\text{isotherm}} < (\int v dp)_{\text{adiabatic}}$

• Volume change smaller for adiabatic comp.

